DATA ANALYTICS SYLLABUS



Module 1: Introduction To Data Analytics

- Introduction to Data Analytics and its importance
- Role, scope and career opportunities
- Types of analytics: Descriptive, Diagnostic, Predictive and Prescriptive
- Data analytics lifecycle
- Understanding structured and unstructured data
- Applications in business, finance, healthcare, marketing and e-commerce

Module 2: Basic To Advanced Excel

- Excel interface, worksheets, formatting and data entry
- Basic formulas and functions
- IF, SUMIF, COUNTIF, SUMIFS and COUNTIFS
- Lookup functions: VLOOKUP, HLOOKUP, XLOOKUP and INDEX-MATCH
- Sorting, filtering, data validation and conditional formatting
- Text, date and logical functions
- Pivot Tables and Pivot Charts
- Charts, reports and interactive dashboards
- What-If Analysis
- Power Query for data cleaning and transformation
- AI-assisted Excel analysis and automation
- **Practical:** Sales reports, employee analysis and Excel dashboards.

Module 3: SQL and Advanced SQL

- Introduction to databases and RDBMS
- Tables, records, primary keys and foreign keys
- SQL data types and constraints
- DDL, DML, DQL and transaction basics
- SELECT, WHERE, DISTINCT, ORDER BY and LIMIT
- Aggregate functions and GROUP BY
- HAVING clause
- Joins: INNER, LEFT, RIGHT and FULL OUTER JOIN
- Subqueries and Common Table Expressions (CTEs)
- CASE statements and NULL handling
- Window functions: ROW_NUMBER, RANK, DENSE_RANK, LEAD and LAG
- Views and stored procedures
- Indexes and basic query optimization
- Hands-on practice using MySQL or PostgreSQL
- **Practical:** Customer analysis, sales analysis and business database queries.

Module 4: Statistics and Probability

- Mean, median, mode, range and percentiles
- Variance and standard deviation
- Basic probability and probability distributions
- Normal, binomial and Poisson distributions
- Population, sample and sampling techniques
- Correlation and covariance
- Introduction to linear regression
- Hypothesis testing and p-values
- t-test, ANOVA and Chi-square test
- Understanding bias, outliers and statistical significance
- **Practical:** Statistical analysis of sales, customer and business datasets.

Module 5: Python Programming — Basic to Advanced

- Introduction to Python and its applications
- Variables, data types and operators
- Input/output and type conversion
- Conditional statements and loops
- Strings and string manipulation
- Lists, tuples, sets and dictionaries
- Functions, arguments, lambda functions and recursion
- List comprehensions
- Modules, packages and virtual environments
- File handling: TXT, CSV and JSON
- Exception handling
- Object-oriented programming fundamentals
- Iterators, generators and decorators
- Working with APIs using the requests library

Module 6: Python Dsa and Problem-Solving

- Time and space complexity fundamentals
- Searching and sorting algorithms
- Linear search and binary search
- Bubble, selection and insertion sort
- Stack and queue concepts
- Linked list fundamentals
- Dictionaries and hash-based problem-solving
- Recursion and backtracking basics
- Two-pointer and sliding-window techniques

- Basic trees and graph concepts
- Python coding exercises and interview problems
- **Practical:** Regular coding exercises and problem-solving sessions.

Module 7: Python For Data Analytics

- NumPy: arrays, indexing, slicing and broadcasting
- Pandas: Series and DataFrames
- Importing data from CSV, Excel, JSON and databases
- Data cleaning and preprocessing
- Handling missing values and duplicate records
- Data transformation, merging and grouping
- Exploratory Data Analysis (EDA)
- Matplotlib and Seaborn visualizations
- Outlier detection and data quality checks
- Excel automation using OpenPyXL
- Working with real-world datasets
- **Practical:** End-to-end data cleaning, analysis and visualization using Python.

Module 8: Microsoft POWER BI

- Introduction to Business Intelligence
- Power BI Desktop interface
- Connecting Excel, CSV, SQL and other data sources
- Data cleaning using Power Query
- Relationships and data modeling
- Star schema fundamentals
- Introduction to DAX
- Calculated columns and measures
- CALCULATE, FILTER, SUMX and time intelligence basics
- Charts, slicers, filters and drill-through
- Interactive dashboards and reports
- Publishing and sharing reports using Power BI Service
- Refresh, security and dashboard performance basics
- AI-assisted analysis and natural-language queries, where available
- **Practical:** Sales dashboard, financial report and business performance dashboard.

Module 9: TABLEAU

- Introduction to Tableau and data visualization
- Connecting to Excel, CSV and SQL data sources
- Data preparation and cleaning

- Dimensions, measures and data types
- Basic charts and visual analytics
- Filters, groups, sets and parameters
- Calculated fields and table calculations
- Data joins, relationships and blending
- Level of Detail (LOD) expressions
- Interactive dashboards and stories
- Tableau Prep fundamentals
- Publishing and sharing dashboards using available Tableau services
- **Practical:** Customer, sales and regional performance dashboards.

Module 10: AI Tools For Data Analytics

- Introduction to AI and Generative AI in analytics
- Prompt engineering for data analysis
- Using ChatGPT and other AI assistants to understand datasets
- AI-assisted Python and SQL query generation
- Debugging and explaining code using AI
- AI-assisted data cleaning and report writing
- Natural-language analysis and visualization
- Introduction to forecasting and anomaly detection
- Understanding predictive analytics and machine learning
- Responsible AI, data privacy and verification of AI-generated results

Module 11: Industry Projects and Career Preparation

- Git and GitHub fundamentals
- Version control and project documentation
- Resume preparation and LinkedIn profile guidance
- Technical interview preparation
- SQL and Python interview questions
- Power BI and Tableau practical assessments
- Mock interviews and project presentation

SUGGESTED PROJECTS

- Excel: Sales performance report and dashboard
- SQL: Customer and product sales analysis
- Python: Data cleaning and exploratory data analysis
- Power BI: Interactive business intelligence dashboard
- Tableau: Customer and regional performance visualization
- Capstone Project: End-to-end business data analytics using Python, SQL and a BI dashboard